

Homework 1: Multivariable Regression Analysis

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1 Research Question

Objective: To evaluate the association between intraventricular hemorrhage (IVH) and neonatal mortality among very low birth weight infants, adjusting for potential confounders.

- **Dataset:** hw_vlbw (Very Low Birth Weight Infants, N=514)
 - **Exposure:** IVH (Intraventricular Hemorrhage)
 - **Outcome:** Death during hospitalization
 - **Study Design:** Retrospective cohort study
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2 Methods

2.1 1. Data Source

- **Sample size:** 514 very low birth weight infants
- **No missing data** on key variables
- **Mortality rate:** 19.3% (99/514)
- **IVH prevalence:** 16.3% (84/514)

2.2 2. Variable Definitions

Primary Variables: - Exposure: IVH (0=No, 1=Yes) - Outcome: DEAD (0=Alive, 1=Dead)

Covariates Considered: - Pneumothorax (pneumo): 0=No, 1=Yes - Delivery mode (delivery_new): 1=C-Section, 2=Vaginal - Multiple gestation (twin): 0=No, 1=Yes - Birth location (inout_new): 1=Born at Duke, 2=Transport - Birth weight (bwt): Continuous (grams) and categorical - bwt_cat: 1=<1000g, 2=1000-1299g, 3=1300-1599g - Gestational age (gest): Continuous (weeks) and categorical - gest_cat: 1=<28 weeks, 2=28-31 weeks, 3= 32 weeks

2.3 3. Statistical Analysis

Step 1: Descriptive Statistics - Baseline characteristics stratified by IVH status - Chi-square tests for categorical variables - t-tests and Wilcoxon rank-sum tests for continuous variables

Step 2: Unadjusted Association - Logistic regression for odds ratio - Poisson regression with log link for risk ratio - 2×2 contingency tables

Step 3: Covariate Selection (Purposeful Selection Method) - Univariate screening: $p < 0.25$ threshold for candidate variables (Hosmer, Lemeshow, &

Sturdivant) - Variables selected: IVH, pneumo, inout_new, bwt_cat, gest_cat
 - Variables excluded: delivery_new (p=0.40), twin (p=0.66)

Step 4: Multivariable Model Building - Initial model with all candidate variables - Sequential removal of non-significant variables (p > 0.10, per purposeful selection workflow) - Retention criteria: removal changes IVH beta by <20% (change-in-estimate criterion for confounding assessment) - Variables removed: twin (p=0.40), delivery_new (p=0.62), inout_new (p=0.93)

Step 5: Final Model - Adjusted for: pneumothorax, birth weight category, gestational age category - Both logistic (OR) and Poisson (RR) regression models estimated

3 Results

3.1 Table 1: Baseline Characteristics by IVH Status

Characteristic	No IVH (n=430)	IVH (n=84)	p-value
Demographics			
Birth weight (g), mean±SD	1101.4±246.0	964.1±266.7	<0.001
Gestational age (wks), mean±SD	29.1±2.3	27.4±2.2	<0.001
Birth Weight Category, n (%)			0.004
<1000g	146 (34.0%)	43 (51.2%)	
1000-1299g	170 (39.5%)	30 (35.7%)	
1300-1599g	114 (26.5%)	11 (13.1%)	
Gestational Age Category, n (%)			<0.001
<28 weeks	114 (26.5%)	40 (47.6%)	
28-31 weeks	255 (59.3%)	42 (50.0%)	
32 weeks	61 (14.2%)	2 (2.4%)	
Complications, n (%)			
Pneumothorax	73 (17.0%)	34 (40.5%)	<0.001
Multiple gestation	100 (23.3%)	7 (8.3%)	0.002
Delivery & Location, n (%)			
C-Section delivery	221 (51.4%)	27 (32.1%)	0.001
Born at Duke	366 (85.1%)	54 (64.3%)	<0.001
Outcome			
Death	58 (13.5%)	41 (48.8%)	<0.001

Key Findings: - Infants with IVH had significantly lower birth weight (137g

difference, $p < 0.001$) - IVH was more common in extremely premature infants (<28 weeks: 47.6% vs 26.5%) - IVH co-occurred frequently with pneumothorax (40.5% vs 17.0%, $p < 0.001$) - Mortality was 3.6 times higher in IVH group (48.8% vs 13.5%)

3.2 Table 2: Mortality by Baseline Characteristics

Characteristic	Alive (n=415)	Dead (n=99)	Mortality Rate	p-value
Exposure				
No IVH	372 (89.6%)	58 (58.6%)	13.5%	<0.001
IVH	43 (10.4%)	41 (41.4%)	48.8%	
Complications				
No pneumothorax	359 (86.5%)	48 (48.5%)	11.8%	<0.001
Pneumothorax	56 (13.5%)	51 (51.5%)	47.7%	
Birth Weight				<0.001
<1000g	122 (29.4%)	67 (67.7%)	35.5%	
1000-1299g	173 (41.7%)	27 (27.3%)	13.5%	
1300-1599g	120 (28.9%)	5 (5.1%)	4.0%	
Gestational Age				<0.001
<28 weeks	93 (22.4%)	61 (61.6%)	39.6%	
28-31 weeks	261 (62.9%)	36 (36.4%)	12.1%	
32 weeks	61 (14.7%)	2 (2.0%)	3.2%	

Interpretation of Table 2: - Mortality was substantially higher among infants with IVH than without IVH (48.8% vs 13.5%), showing a strong crude exposure-outcome gradient. - Mortality was also much higher among infants with pneumothorax (47.7% vs 11.8%), indicating severe respiratory complications are strongly related to death. - A clear dose-response pattern was present for birth weight: mortality decreased from 35.5% (<1000g) to 13.5% (1000-1299g) to 4.0% (1300-1599g). - A similar gradient was observed by gestational age: 39.6% (<28 weeks), 12.1% (28-31 weeks), and 3.2% (≥32 weeks), supporting prematurity as a major mortality driver. - Because IVH is associated with these strong mortality predictors (Table 1), confounding is plausible and multivariable adjustment is required.

3.3 Table 3: Unadjusted and Adjusted Association Between IVH and Mortality

Model	Measure	Estimate	95% CI	p-value
Unadjusted				
Logistic	Odds Ratio	6.116	3.674 - 10.179	<0.001
Poisson	Risk Ratio	3.619	2.426 - 5.398	<0.001
Adjusted¹				
Logistic	Odds Ratio	4.149	2.291 - 7.516	<0.001
Poisson	Risk Ratio	2.059	1.351 - 3.137	0.001

¹ Adjusted for pneumothorax, birth weight category, and gestational age category

Interpretation: - After adjusting for key confounders, IVH **increased the odds of death by 4.1-fold** (95% CI: 2.3-7.5) - We emphasize RR over OR for effect communication because death is not a rare outcome in this cohort ($99/514 = 19.3\%$); with common outcomes, OR can overstate the magnitude of association relative to RR. - Using the more appropriate risk ratio for this context, IVH **doubled the risk of death** (RR=2.1, 95% CI: 1.4-3.1) - Adjustment attenuated the effect by 32% (OR: 6.1→4.1), indicating substantial but incomplete confounding

3.4 Table 4: Final Multivariable Model Results**3.4.1 A. Logistic Regression (Odds Ratios)**

Variable	Odds Ratio	95% CI	p-value
IVH (Yes vs No)	4.149	2.291 - 7.516	<0.001
Pneumothorax (Yes vs No)	6.185	3.486 - 10.975	<0.001
Birth Weight			0.0002
<1000g vs 1300-1599g	8.902	2.858 - 27.729	0.0002
1000-1299g vs 1300-1599g	3.216	1.113 - 9.293	0.031
Gestational Age			0.151
<28 weeks vs 32 weeks	3.691	0.749 - 18.177	0.108
28-31 weeks vs 32 weeks	2.239	0.484 - 10.352	0.302

Model Performance: - C-statistic: 0.853 (excellent discrimination) - AIC: 373.1 - -2 Log Likelihood: 359.1

3.4.2 B. Poisson Regression (Risk Ratios)

Variable	Risk Ratio	95% CI	p-value
IVH (Yes vs No)	2.059	1.351 - 3.137	0.001
Pneumothorax (Yes vs No)	2.821	1.850 - 4.300	<0.001
Birth Weight			0.004
<1000g vs 1300-1599g	4.861	1.779 - 13.283	0.002
1000-1299g vs 1300-1599g	2.692	1.025 - 7.069	0.044
Gestational Age			0.244
<28 weeks vs 32 weeks	3.143	0.710 - 13.923	0.132
28-31 weeks vs 32 weeks	2.333	0.555 - 9.809	0.248

3.5 Sensitivity Analysis

A simpler model including only IVH, pneumothorax, and birth weight (without gestational age) was also tested:

Parameter	IVH OR (Full Model)	IVH OR (Simple Model)	% Change
Beta coefficient	1.4230	1.4750	+3.7%
Odds Ratio	4.149	4.371	+5.3%
C-statistic	0.853	0.842	-1.3%

Conclusion: The IVH effect estimate is robust across model specifications (<5% change), supporting the validity of our findings. The full model with gestational age is retained as the primary analysis due to slightly better discrimination and theoretical considerations.

4 Discussion

4.1 Main Findings

1. **Strong Independent Effect:** After controlling for prematurity markers (birth weight, gestational age) and complications (pneumothorax), IVH re-

mained strongly associated with neonatal mortality (adjusted OR=4.1, 95% CI: 2.3-7.5).

2. **Substantial Confounding:** The crude association (OR=6.1) was attenuated by 32% after adjustment, indicating that the relationship between IVH and mortality is partially explained by shared risk factors related to prematurity and illness severity.
3. **Pneumothorax as Key Predictor:** Pneumothorax showed the strongest association with mortality (OR=6.2), suggesting that respiratory complications are critical determinants of outcomes in this population.
4. **Dose-Response Relationship:** A clear gradient was observed for birth weight, with the lowest weight category (<1000g) having nearly 9-fold increased odds of death compared to the reference group (1300-1599g).
5. **Gestational Age Effect:** While gestational age categories showed non-significant associations in the multivariable model ($p=0.15$), this likely reflects collinearity with birth weight rather than lack of clinical importance.

4.2 Why RR is Preferred for Primary Interpretation

- The outcome (in-hospital death) occurred in 19.3% of infants, which is above the usual “rare outcome” range where OR approximates RR.
- OR is mathematically valid for logistic models but tends to exaggerate effect size when outcomes are common.
- RR is more directly interpretable for cohort-style risk communication (“times the risk”) and better aligns with clinical/public-health interpretation.
- Therefore, OR and RR are both reported for completeness, but RR is used as the primary effect measure in interpretation.

4.3 Clinical Implications

- IVH represents a **major independent risk factor** for mortality in very low birth weight infants
- Early detection and management of IVH should be prioritized
- Co-occurrence of IVH and pneumothorax identifies an extremely high-risk subgroup (likely >50% mortality)
- Prevention strategies targeting the most premature infants (<28 weeks, <1000g) may have the greatest impact on reducing mortality

4.4 Strengths

- Complete data with no missing values

- Appropriate statistical methods (purposeful selection, both OR and RR reported)
- Robust effect estimates across model specifications
- Excellent model discrimination (c-statistic=0.853)

4.5 Limitations

1. **Collinearity:** High correlation between birth weight and gestational age limits ability to disentangle their independent effects
2. **Residual Confounding:** Unmeasured factors (e.g., severity of IVH, ventilator settings) may still confound the association
3. **Sample Size:** Wide confidence intervals for some estimates, particularly gestational age categories
4. **Temporal Ambiguity:** Cannot determine if some complications (e.g., pneumothorax) occurred before or after IVH

5 Conclusions

Intraventricular hemorrhage is a strong, independent predictor of neonatal mortality among very low birth weight infants. After adjusting for pneumothorax, birth weight, and gestational age, infants with IVH had **4.1 times higher odds** (OR=4.1, 95% CI: 2.3-7.5) and **2.1 times higher risk** (RR=2.1, 95% CI: 1.4-3.1) of death compared to those without IVH. These findings underscore the critical importance of preventing and managing IVH in this vulnerable population.

6 References

1. Hosmer DW, Lemeshow S, Sturdivant RX. *Applied Logistic Regression*. 3rd ed. New York: Wiley; 2013. (Purposeful selection workflow, including $p < 0.25$ screening and iterative model reduction.)
2. Mickey RM, Greenland S. The impact of confounder selection criteria on effect estimation. *Am J Epidemiol*. 1989;129(1):125-137. (Rationale for change-in-estimate confounder assessment.)
3. Zhang J, Yu KF. What's the relative risk? A method of correcting the odds ratio in cohort studies of common outcomes. *JAMA*. 1998;280(19):1690-1691. (Why OR diverges from RR when outcome is common.)

4. Zou G. A modified Poisson regression approach to prospective studies with binary data. *Am J Epidemiol.* 2004;159(7):702-706. (RR estimation using Poisson model with log link for binary outcomes.)

Software: - SAS 9.4 (SAS Institute, Cary, NC) - Procedures: PROC LOGISTIC, PROC GENMOD, PROC FREQ, PROC MEANS

7 Appendix: Variable Coding

IVH: 0 = No IVH, 1 = IVH present
 DEAD: 0 = Alive at discharge, 1 = Dead
 pneumo: 0 = No pneumothorax, 1 = Pneumothorax
 bwt_cat: 1 = <1000g, 2 = 1000-1299g, 3 = 1300-1599g (reference)
 gest_cat: 1 = <28 weeks, 2 = 28-31 weeks, 3 = 32 weeks (reference)
 delivery_new: 1 = C-Section (reference), 2 = Vaginal
 twin: 0 = Singleton (reference), 1 = Multiple gestation
 inout_new: 1 = Born at Duke (reference), 2 = Transport

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Analysis Code: HW1_Analysis.sas